

Analysis

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1. Introduction

This project focuses on developing a **secure notepad-style web application** where users can safely write, save, and manage notes.

The main objective is to apply **practical cybersecurity principles** in a functional and realistic environment, using **Flask (Python)** for the backend, **HTML/CSS** for the frontend, and **SQLite** for database management. The application will be deployed using the **NetLab infrastructure** and later potentially hosted on a **Raspberry Pi**, adding a layer of network configuration and server management experience.

2. Educational Objective

The educational goal of this project is to deepen understanding and practical skills in:

- **Encryption** algorithms.
- Authentication systems using **hashing**.
- Secure data storage and communication protocols (**SSL/TLS**).
- **Hosting** and deploying secure web applications on managed infrastructure.

By completing this project, I aim to bridge theoretical cybersecurity knowledge with real-world application - gaining both **security implementation** and **network administration** experience.

3. Main Questions

- How can user data be stored securely while maintaining accessibility and usability?
- How can SSL/TLS be configured to guarantee secure client-server communication?
- What best practices should be applied when hosting a web app on an external server?

4. Functional Requirements

- Users can create an account.
- User credentials are hashed and stored securely.
- Users can write, save, and edit notes.
- The app must use HTTPS and secure certificate configuration.

4.1 MoSCoW Prioritization

Must Have:

- Account creation, login, and creation of notes.
- Secure storage of information.
- HTTPS and SSL/TLS configuration.

Should Have:

- Multi-user support with role separation.
- Interface for managing notes intuitively.

Could Have:

- AES256 encryption for key generation.

Won't Have:

- Animations or unnecessary visual effects.

4.2 Agile Methodology

Development follows an **Agile** approach, focusing on iterative cycles, constant testing, and reflection. Tasks are organized into sprints with weekly deliverables, ensuring steady progress while maintaining flexibility for adjustments and improvements.

User Stories

- As a user, I want to create an account so I can store my notes securely.
- As a user, I want my notes to be encrypted automatically so that no one else can read them.
- As a user, I want to log in and access my notes easily without compromising security.
- As a developer, I want to understand how encryption and authentication can coexist efficiently.

Agile Practices

- Daily self-checks: Track progress against functional requirements.
- Sprint reviews: Demo prototypes to peers, teachers or self-test.
- Retrospectives: Reflect on coding challenges and workflow efficiency.

5. Sprint Planning

- **Sprint 1** (Week 1): Define project criteria and research encryption algorithms; Conduct feedback sessions with teachers and peers to refine direction.
- **Sprint 2** (Week 2): Draft system architecture, decide what encryption will be used and ask for advice and feedback.
- **Sprint 3** (Week 3): Build the backend and UI/UX, prepare hosting services, ask for feedback and fix bugs if any.
- **Sprint 4** (Week 4): Perform testing, work on hosting the web application, set up HTTPS, put the accumulated evidence in portfolio and ask for feedback.

6. Stakeholder Analysis

Primary Stakeholders:

- **David Hadzhiivanov** – developer, system administrator, and primary decision-maker.
- **Fontys Teachers and Peers** – evaluators and feedback providers for both technical and academic progress.

Secondary Stakeholders:

- Future users or testers who may interact with the prototype for validation purposes.

7. Precognitions

- The project will rely on Python, and SQLite as core technologies.
- Carried out with school resources and no external cost.
- Open source, allowing future improvement.
- Familiarity with web development principles (HTML, CSS basics).
- Network infrastructure (NetLab / Raspberry Pi) will be accessible and stable.
- Time limited to 4 weeks.

8. System Constraints

Hardware:

- Development computer.
- Raspberry Pi for deployment.

Software:

- Python 3.x
- Flask
- SQLite

Network:

- Reliable connection for hosting.
- HTTPS support.

9. Risk Analysis

Risk	Probability	Impact	Mitigation
Chosen encryption algorithm may not meet desired security standards.	Medium	High	Maintain secure algorithms that have wide support and documentation.
Time constraints may limit UI/UX polish.	High	Low	Keep the front end simple and intuitive.
SSL/TLS misconfiguration.	Low	Medium	Use verified certificates and proper configuration settings.

Server failure.	Medium	High	If Fontys hosting doesn't work keep raspberry pi as backup.
Database breach.	Medium	High	Use strong hashing algorithms and test login flow thoroughly.

10. Conclusion

This analysis defines the foundation of the **Secure Notepad Web Application** - a cybersecurity-focused project emphasizing encryption, secure data handling, and deployment. Through iterative Agile development, the project will balance theory with practice, improving both programming discipline and security understanding.